Does the combinatorial finding support the display inference?

Adversarial review, lens: **generalisation** — the finding is granted in full; the target is the inference drawn from it. 20 October 2026. All figures below were regenerated from `runs/` and from new runs made for this review.

> **Verdict: SERIOUSLY_WEAKENED.**
> The finding reproduces exactly. The inference does not follow. On the project's own geometry metric, its own default macrocycle library sits at the *opposite* end of the axis from the two landscapes where the policy helps, and the gain decays as a clean power law in pool size (exponent -0.71, $R^2 = 0.990$) along the one axis that actually points toward display.

1. The numbers reproduce

Recomputed from the delay-0 half of the 628 run files in `runs/method/` and `runs/targets/`, paired per pool against `random` at the final 128-label checkpoint, delay 0. Every cell matches the claim to three decimals.

landscape	kind	n pools	greedy	sd	wins	ucb	sd	wins
spg1_olson_igg_matched	combinatorial	10	+0.242	0.218	8/10	+0.304	0.182	9/10
yap1_ww_peptide	combinatorial	18	+0.263	0.164	17/18	+0.246	0.165	17/18
ha_h3	single-sub	10	+0.104	0.085	8/10	+0.050	0.093	6/10
mpro	single-sub	10	+0.065	0.152	7/10	+0.038	0.120	5/10
ha_h1	single-sub	10	−0.004	0.118	6/10	+0.046	0.107	7/10
spike_rbd_ace2	single-sub	6	+0.030	0.136	3/6	−0.037	0.091	2/6

Table 1. Paired top-5% recall gain over random. Reproduced, not taken on trust.

Granted. The rest of this review attacks only the sentence that follows it: *“an mRNA display library is combinatorial, so the regime where the policy helps is the regime the real application lives in.”*

2. The fatal objection: “combinatorial” is doing equivocal work

The word covers two unrelated geometries, and the project's two winning landscapes are the wrong one.

What the landscapes actually contain. `spg1_olson_igg` is 535,917 *double* mutants plus 1,045 singles — nothing deeper. `yap1_ww_peptide` is 9,713 doubles plus 362 singles. Every candidate is wild-type $\pm$ 2 substitutions on a shared, evolved, high-affinity background. These are not “a randomised region drawn from an alphabet.” They are perturbation scans that happen to perturb two sites instead of one.

Measured on the project's own metric. `binding.py:distances()` scores candidates by mean normalised Hamming distance. Running it over each pool, and over a library sampled from the project's own `chemistry.library_spec()`:

pool / library	feature columns	mean pairwise distance	policy gain?
spg1_olson_igg_matched	55	**0.071**	large (+0.24 / +0.30)
yap1_ww_peptide	34	**0.111**	large (+0.26 / +0.25)
SynX default macrocycle library (chemistry.library_spec)	10	**0.736**	— unmeasured —
mpro	2	0.974	small (+0.07 / +0.04)
ha_h1 / ha_h3 / spike_rbd_ace2	2	0.973–0.974	small or zero

Table 2. Mean normalised Hamming distance between pool members, the exact quantity `distances()` computes and the `diverse` policy maximises.

On the project's own representation, its own library is **ten times closer to the single-substitution scans than to the two landscapes it is being used to argue from**. The inference asks the reader to accept the macrocycle library as the combinatorial case. The code says it is the dispersed case.

Sampled with `sample_library(library_spec(), 4000, rng(5))`; restricted to a single length the figure is 0.768. The spec is a 4–8 residue randomised region over a 25-block alphabet, theoretical diversity 1.589×10^{11}.

3. The gain decays as a power law along the only axis that points at display

The one dimension the project can genuinely vary toward the display regime is pool size at a fixed label budget. New runs, same `replay()`, same 64-tree forest, 128 labels, top-5% threshold held constant, 10 disjoint pools at every size:

pool	labelled fraction	random	greedy	× over random	headroom	fraction of headroom	normalised gain
512	25.0%	0.260	0.503	1.93×	4×	0.48	**0.328**
1,024	12.5%	0.117	0.299	2.56×	8×	0.32	0.206
2,048	6.25%	0.060	0.199	3.31×	16×	0.21	0.148
4,096	3.12%	0.028	0.116	4.10×	32×	0.13	0.090
8,192	1.56%	0.015	0.063	4.33×	64×	0.068	0.050
16,384	0.78%	0.007	0.034	4.86×	128×	**0.038**	**0.028**

Table 3. spg1_olson_igg_matched. Budget and threshold fixed; only the pool grows. Normalised gain is `(policy − random)/(1 − random)`, the quantity `summarize_method()` already uses to compare unequal geometries.

Normalised gain $\propto$ pool_size$^{-0.71}$, $R^2 = 0.990$ over a 32× range. Note the trap: the raw *multiple over random* rises from 1.93× to 4.86× and would read as the policy improving. It is not. The multiple is bounded by 1/labelled-fraction, and as a fraction of that ceiling the policy falls from 0.48 to 0.038 — a 13-fold collapse. This is the same shape as the "8× better than random" artefact the team was burned by, pointing the other way.

Extrapolating the fit is not a prediction, but it bounds the argument: at the project's own library size of 1.6×10^{11} the fitted normalised gain is 3×10^{-7}. **Whatever justifies running an acquisition policy on a**

display library, it is not this curve. yap1 behaves the same way over the range its 10,075 rows allow: +0.322 at 512, +0.202 at 1,024, +0.168 at 2,048.

4. Five further differences, each of which alone breaks the mapping

4.1 “Request this candidate's measurement” is not an action in display

This is the decisive structural difference and it is not a matter of degree. A RaPID round incubates the *entire* library against immobilised target, washes, elutes, amplifies, and sequences the survivors. The experimenter chooses library composition, stringency, competitor, negative selections and round count. The experimenter never chooses which 8 members to assay. The policy implemented in `pick_batch()` — select 8 of 512, read their scores, refit, repeat 15 times — has no counterpart in a selection round. It has a counterpart only in *deconvolution*: resynthesising and individually measuring a few dozen clones drawn from the post-selection survivor list. That is a genuine 10^2–10^5 candidate problem with per-candidate readout, and it is where this machinery belongs.

4.2 The “enrichment readout” in the code is not an enrichment readout

`binding.py:enrichment()` is documented as “the readout shape a display selection round actually produces.” It is not. Checked across all 3,140 recorded checkpoints, the maximum deviation from `top_recall × pool_size / labels` is exactly $\mathbf{0.00 \times 10^0}$. At a fixed budget it is an algebraic rescaling of recall and carries zero additional information. It contains no count statistics, no sequencing depth, no dropout, no PCR or translation bias, no bead or linker artefact, and no round-over-round amplification dynamics — which is to say, none of the things that make an enrichment ratio hard to act on. The project should not treat this function as evidence that the display readout has been modelled.

4.3 Batch structure

Sixteen sequential batches of 8 with complete, noiseless feedback after each, versus 4–6 selection rounds applied to the whole library at once with feedback available only as sequence counts at the end. The project's own delay-3 arm is the right instinct but the wrong magnitude: it models three decision rounds of latency, not a readout that never resolves per candidate.

4.4 Chemistry, and a representation that cannot hold it

Every one of the six landscapes is 20 canonical L-amino acids on a folded domain of 34–566 residues with a fixed backbone. The library spec is a chloroacetyl initiator, a thioether macrocycle, N-methyl and D-amino acid blocks, and a *variable-length* 4–8 residue region. `sample_library()` returns members of 6–10 blocks; the feature matrices `binding.py` consumes are fixed-width. The registry carries no SMILES, CCD code, stereochemistry descriptor or molecular weight for any non-canonical block. The manifests already state this; the inference quietly steps over it.

4.5 The interface is mutated on the wrong side

Both winning landscapes mutate the *protein* side of an evolved interface against a fixed natural partner. So does `dlg4_pdz3_peptide`. No landscape on disk mutates the binder. The two tasks are not variants of each other: substitution tolerance asks how far an optimised interface can be pushed before it degrades, and the answer is dominated by additive site effects measured against a working reference. De novo binder design asks which of many unrelated sequences creates an interface at all, where almost every member is a non-

binder and there is no reference to measure against. The finding is about the first; the application is the second.

5. A confound that also damages the finding itself

This review was asked to grant the finding, but one check bears directly on whether the "regime" is a property of the landscape or of the surrogate. Fitting a plain ridge regression on the same one-hot features, from 128 random labels, and measuring held-out Spearman inside each pool:

landscape	additive ridge ρ	random forest ρ	ratio
spg1_olson_igg_matched	0.706	0.658	1.07×
yap1_ww_peptide	0.441	0.341	1.29×
mpro	0.298	0.103	**2.89×**
spike_rbd_ace2	0.273	0.055	**4.96×**
ha_h1	0.191	0.092	2.08×
ha_h3	0.162	0.056	2.89×

Table 4. The fixed 64-tree forest is beaten by a linear model on every landscape, and beaten badly on the single-substitution scans.

Replacing the surrogate inside the acquisition loop, with the same seeds, the same shared initial 8 labels and the same paired random baseline:

landscape	n	gain, RF greedy	gain, ridge greedy	gain, ridge UCB
spg1_olson_igg_matched	10	+0.242	+0.514	+0.456
yap1_ww_peptide	18	+0.263	+0.329	+0.318
mpro	10	+0.065	+0.150	+0.154
ha_h3	10	+0.104	+0.096	+0.050
ha_h1	10	−0.004	+0.073	+0.100
spike_rbd_ace2	6	+0.030	−0.006	+0.005

Table 5. Paired gain over random with a ridge surrogate instead of the 64-tree forest.

Two consequences. First, `ha_h1`'s negative gain and `mpro`'s small gain are substantially artefacts of a surrogate that suits near-wild-type doubles and does not suit a (position, residue) encoding; three of the four single-substitution landscapes become clearly positive. Second, that a ridge regression *beats* the forest on the combinatorial landscapes identifies the mechanism: the exploitable structure is **additive site effects measured against a shared wild-type background**. That is precisely the structure a naive randomised macrocycle library does not have — there is no shared background, and in a 4–8 residue cyclic peptide with no fold, position effects are strongly context-dependent. The mechanism that makes the finding true is the mechanism that makes it untransferable.

The gap does not close — combinatorial gains stay well above single-substitution gains under both surrogates — so the finding survives in weakened form. But "the ranges do not overlap" is a statement about one fixed forest, not a categorical regime boundary.

6. What would have to be true, and the experiment that would test it

For the inference to hold, all four of these would have to be true. None is currently established:

1. The gain survives at a labelled fraction near the application's, not at 25%. Section 3 says it does not survive even to 0.78%.
2. The gain survives when candidates are mutually distant rather than clustered around a reference. Untested; Table 2 says the application is at the untested end.
3. The gain survives when the readout is a noisy pooled enrichment ratio rather than a clean per-candidate score. Untested; §4.2 shows the code does not currently model this at all.
4. Per-candidate acquisition is an available action. In selection it is not; in deconvolution it is.

The experiment. A single deep-mutational-scan cannot settle this, because no DMS has the geometry. The test is a landscape with (a) many simultaneously randomised positions and no shared wild-type background, (b) a combinatorially complete or near-complete measured readout, and (c) a peptide-ligand-against-fixed-target structure. Candidates that exist in public data:

- A fully randomised short-peptide binding dataset — e.g. a yeast- or phage-display peptide library with per-variant measured affinity over a randomised 5–8mer, rather than a substitution scan of a folded domain.
- Failing that, the honest minimum: take `spg1_olson_igg` and construct pools whose members are *mutually* distant rather than all 2 substitutions from wild-type, and rerun. If the gain survives dispersion at matched pool size and matched budget, the inference gains its first real support. If it does not, the inference is dead and the project saves itself a Thursday claim it would have to retract.
- Separately, and independently useful: re-run the whole comparison with a counts-based enrichment readout — multinomial sampling at realistic depth, with dropout — in place of the current exact rescaling of recall.

The second of those costs a few hours of the machinery that already exists. It should be run before Thursday.

7. The version of this finding that is directly useful

There is one, and it is worth more than the overreach it replaces, because it describes work the project will actually do:

Claim it for deconvolution and for affinity maturation, not for selection. After a display campaign, the team holds a ranked survivor list and must decide which few dozen macrocycles to resynthesise and measure individually by SPR or IC_{50}. That problem has exactly the geometry the result covers: hundreds to thousands of candidates, a genuine per-candidate measurement request, a small batched budget, a labelled fraction in the percent range, and — after the first hit — candidates clustered a few substitutions from a shared lead. A focused re-library around a confirmed hit is *precisely* the wild-type $\pm$ 1–3 geometry of `spg1_olson_igg`, mean pairwise distance and all.

Stated that way the result is defensible, it is testable before any macrocycle data arrives, and it names a decision the platform really has to make. Stated as “display libraries are combinatorial, so this is our regime,” it is an equivocation on the word *combinatorial* that the project's own `distances()` function refutes in one line.

8. Caveats that must travel with any version of the claim

- The hypothesis was generated by the 19 October runs and tested on data staged on 20 October to test it. `spg1_olson_igg_matched` was created for this purpose and must be reported as a follow-up test, not an independent confirmation. The manifest already says so.
- The combinatorial arm is n = 2 landscapes and one of them is the same protein (GB1) as the original pilot. Effectively n = 1 independent system plus a restaging.
- "Combinatorial" here means doubles on a wild-type background. It does not mean a randomised region.
- Assay endpoints differ across landscapes (IgG binding selection, peptide binding, replication fitness, FRET cleavage). The combinatorial/single-substitution split is perfectly confounded with a binding/fitness split: both high-gain landscapes are binding assays.
- Pool spread is variability across sub-pools of one protein, not biological replication.
- No macrocycle, non-canonical residue, cyclisation or enrichment readout appears in any of this. Nothing here speaks to the chemistry that defines the platform.

Regenerated from `runs/method/` and `runs/targets/` with `./.venv-hpc/bin/python`. New runs for Tables 3–5 used `hpc_pilots.binding.replay` unmodified except for the surrogate, with the project's own pool-construction seed (20261019).